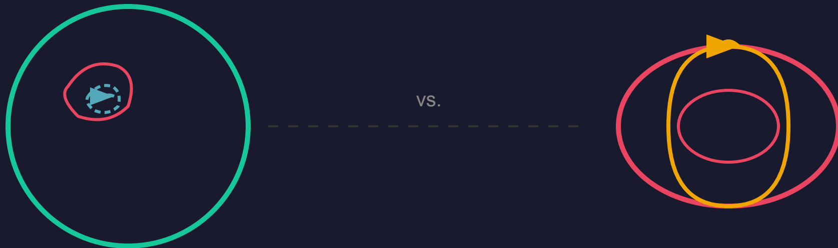
Fundamental Group: Simply Connected Spaces



S^2

Simply Connected

$\pi_1(S^2) = 0$

T^2

NOT Simply Connected

$\pi_1(T^2) = \mathbb{Z} \times \mathbb{Z}$

Key Insight

On S^2, every loop can be continuously shrunk to a point.

On T^2, loops going around the hole cannot be shrunk.

The Poincaré Conjecture: S^3 is the only 3-manifold with this property!